L3 Data Specialist

Module 6: From Data Citizen to Data Specialist

Three-day class-based learning

QA

Learn. To Change.



Session overview



Introductions and overview of the three-day class-based learning.



Engage in practical activities to support module evidence collection.



Recap Module 6 key concept digital learning content.



Provide support and guidance for the successful completion of Module 6.

The KSB framework: Knowledge

- **K4:** Data element formats and how their selection can impact precision, analysis, and communication, for example, integers, floating point numbers and their precision, scientific notation, date formatting as strings.
- **K5:** How to access and extract data from already identified sources.
- **K6:** How to collate and format data in line with organisational standards.
- **K13:** How to combine data from multiple sources. For example, using look ups, copy and paste, and visualisation tools or data blending tools on bespoke systems.

The KSB framework: Knowledge

- **K14:** Understand the capabilities within data analysis tools, for example, spreadsheets or database viewers or digital display screens on bespoke systems for use in answering questions, solving problems, and the potential to use automation for repeated data manipulation.
- **K15:** How to filter details, focusing on information relevant to the data tasks and purpose.
- **K6:** How to collate and format data in line with organisational standards.
- **K16:** Basic statistical methods to extract relevant information from structured and unstructured data, for example, counting rows, calculating the mean and standard deviation of numeric fields, counting words in a document, listing the most common values, calculating percentage contributions or percentage differences between data items.

The KSB framework: Skills

- **S1:** Select and migrate data from already identified sources.
- **S2:** Format and save datasets.
- **S3:** Summarise and explain gathered data.
- **S4:** Combine datasets from multiple sources and present in format appropriate to the task.
- **S5:** Use tools and/or apply basic statistical methods to identify trends and patterns in data.

The KSB framework: Skills

- **S6:** Identify faults and cleanse data for example identifying gaps, duplicate entries, outliers, and unusual variances, including cross-checking across data elements or between data sources.
- **S7:** Audit data results, reviewing a dataset once all sources are combined, to ensure accuracy, completeness, consistency, and traceability from original data.
- **S8:** Demonstrate the different ways of communicating meaning from data in line with audience requirements.
- **S16:** Demonstrate the ability to use different tools and methods to formulate and utilise effective prompt to research, apply, and evaluate data transformation techniques

Schedule: Day 1

AM

- Changes in the digital landscape and emerging technologies
- Activity: Research and report on emerging technologies
- The Internet of Things (IoT)
- IOT impacts on industry

PM

- GenAI uses
- Prompt engineering
- Activity: AI prompts
- Cloud for data analysis
- Activity: Cloud use for organisations

Schedule: Day 2

AM

- EPA methods and structure
- EPA prep: Scenario demonstration practice (part guided)

PM

- Continued: EPA prep: Scenario demonstration practice (part guided)

Schedule: Day 3

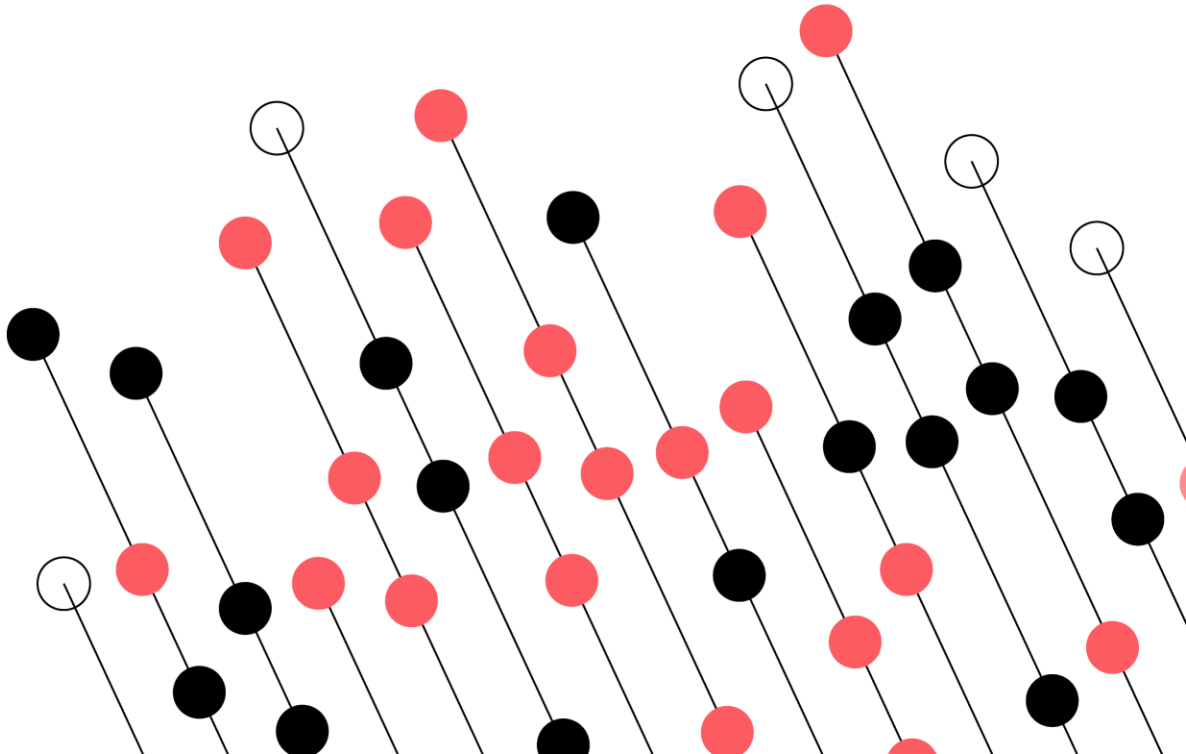
AM

- Paired mock 1: Learner and assessor scenario demonstration practice

PM

- Paired mock 2: Learner and assessor scenario demonstration practice

Day 1

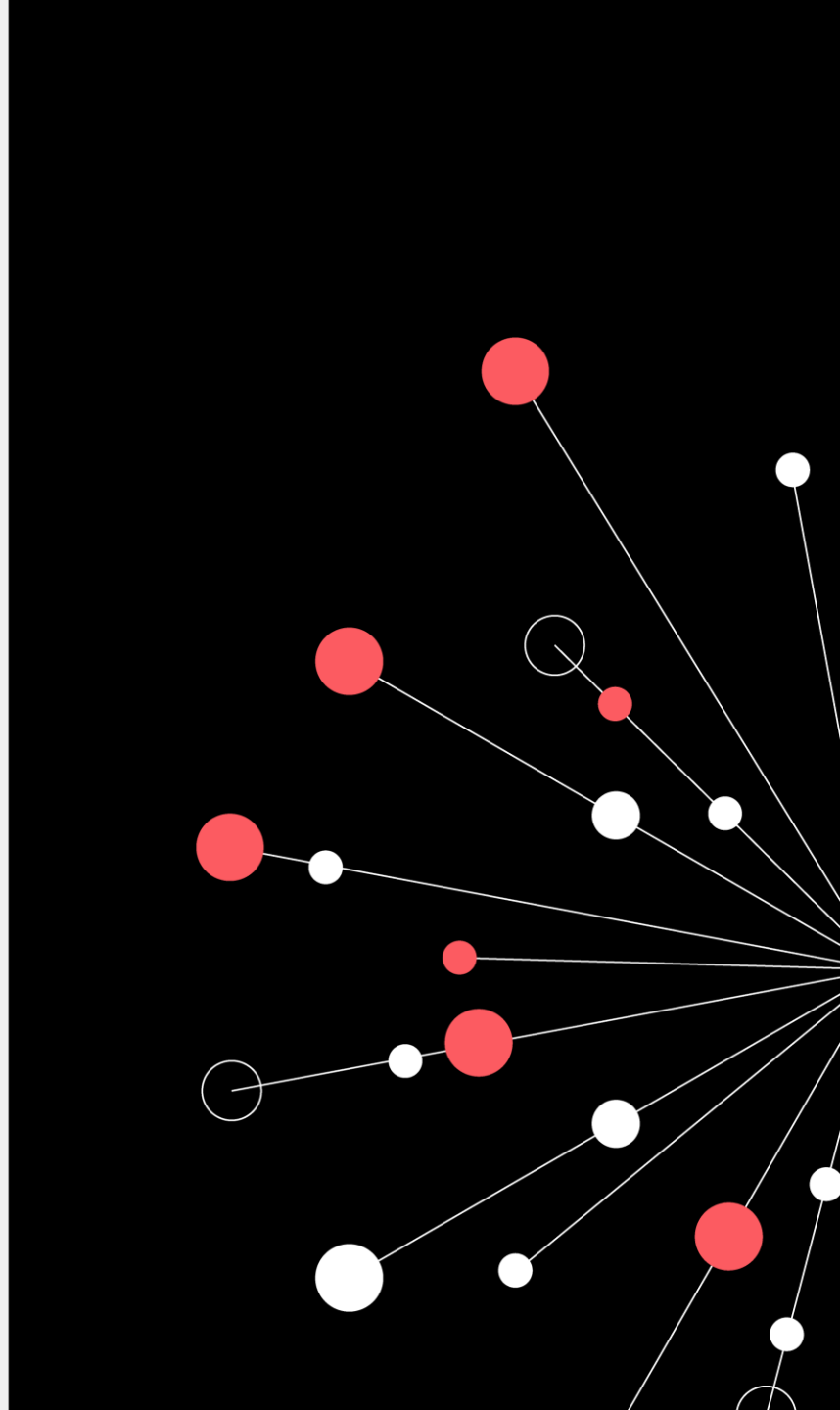


Discussion: Your organisation

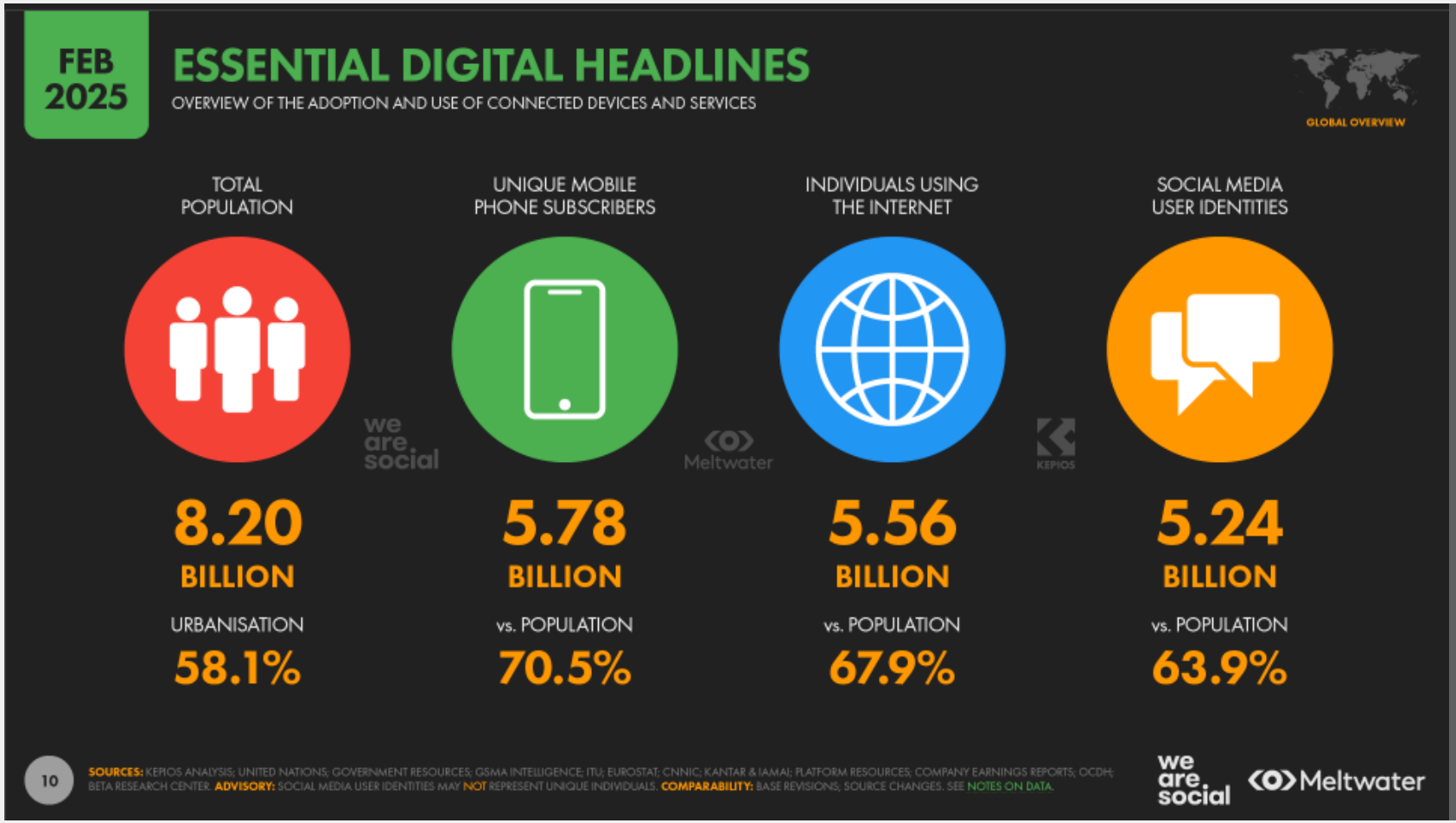
You have **15 minutes** to think and make **brief** notes on the following.

- A short intro to your organisation and what they sell / provide (target market, and so on).
- Any major inadequacies in the organisation's / industry's technological processes.
- A description of technological processes that your organisation could improve / innovate with in the next 1-5 years.

You will then feedback your answers.



The changing digital landscape



Data underpins every digital interaction across the digital landscape, including applications, devices, Internet of Things (IoT), and customer centricity.

Emerging digital trends

The Digital Annual Global Overview Report provides extensive coverage of the latest data insights and social media statistics.

‘Data is the new oil’ — similar reports can support strategic decision-making in organisations.

As part of your apprenticeship, you have been analysing and reporting on data trends.

The annual Digital Global Overview Report will further enrich your knowledge and understanding of digital trends.



Task: Understand the impact of emerging technologies on your organisation / industry

1. Conduct research on digital trends and emerging technologies, specifically linked to your organisation / industry.
2. Using the insights gained from your research, write a report on how digital trends and emerging technologies could:
    a) Help in your organisations' digital strategic planning.
    b) Help improve processes.
    c) Boost business value for the organisation / sector.

Include any relevant figures, quotes, statistics, or graphics from your research.

Suggestions:

- The use and growth of social media platforms, or changes in behaviours because of global events and influences.
- Growing use of AI for communication, customer service, report generation, data summarisation, and so on.
- Customers accessing your products / services in a more effective and efficient manner.
- Mobile device, IOT, internet users, and uses.

Approx 90 mins: 30 mins research, 60 mins report.

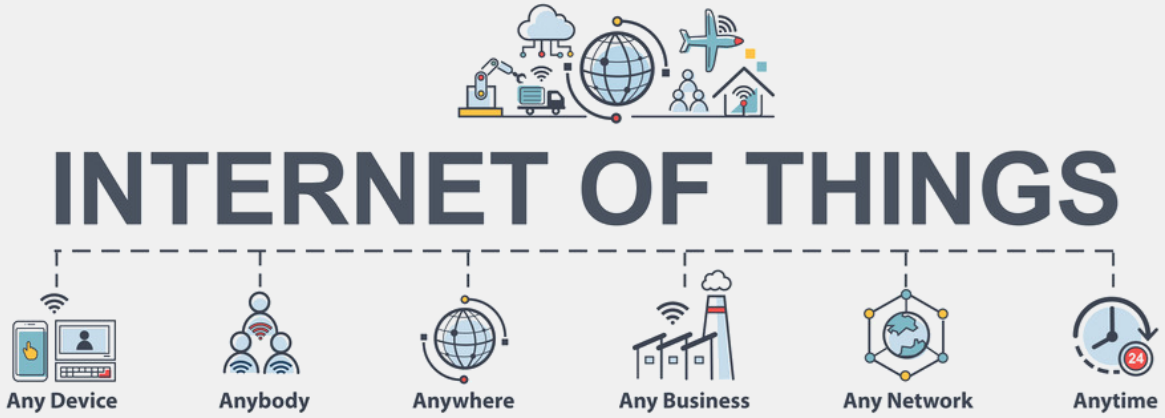
Internet of Things (IoT)

You've visited the Internet of Things in previous modules.

Take 10 minutes to think, research, and make notes into:

- What does IoT describe?
- Examples of IoT.
- How does it form part of your organisation?
- Has the use of IoT changed over the past few years?

Discuss your answers as a class.



Activity: Internet of Things — Impacts on industry

Tasks:

1. Read the article and discuss the following key areas in groups (30 mins). Answer: **how has IoT and automation impacted on the following industry sectors?**
    - Retail and supply management
    - Automotive
    - Agriculture
    - Logistics
    - Healthcare

2. Feedback to the class.



https://easternpeak.com/blog/6-cool-examples-of-internet-of-things-applications-and-how-to-develop-one/

AI

There are two main types:

- Narrow AI (weak).
- General AI (strong).

Key concepts in AI include:

- Machine learning (ML).
- Deep learning.
- Natural language processing (NLP).



AI is used in many applications, for example, recommendation systems, virtual assistants autonomous vehicles, medical diagnosis, robotics, and scientific research.

GenAI (generative AI)

GenAI is a subset of artificial intelligence that focuses on creating new, original content based on patterns learnt from existing data.

Unlike traditional AI, which is typically designed to perform specific tasks like classification or prediction, generative AI can generate text, images, music, videos, and even code that mimics human creativity.

It works by training on large datasets, learning the underlying structure and patterns within that data, and then using this knowledge to produce new, coherent outputs that resemble the original data but are not direct copies. Popular examples include tools like GPT (for text generation), DALL·E (for image generation), and music composition algorithms.

Source: ChatGPT

Leading GenAI models: Text generation

Model	Company	Use
GPT	OpenAI	Multimodal, advanced reasoning, used in ChatGPT
Claude 3 Opus	Anthropic	Safe, transparent, long-context capabilities
Gemini	Google DeepMind	Long memory, deep integration with Google products

Leading GenAI models: Image generation

Model	Company	Use
DALL·E	OpenAI	Integrated with ChatGPT, excellent prompt-following
Midjourney	Midjourney	Artistic, creative image generation
Stable Diffusion XL (SDXL)	Stability AI	Open-source, flexible for customisation and control

Leading GenAI models: Video generation

Model	Company	Use
Sora	OpenAI	Text-to-video model, highly realistic and cinematic
Runway Alpha	Runway	Real-time and stylised video generation
Pika	Pika Labs	Short-form, creative video editing and generation

Leading GenAI models: Music generation

Model	Company	Use
Suno AI	Suno	Text-to-song, including vocals and full tracks
Udio	Udio Labs	High-quality, AI-generated music with lyrics
MusicGen	Meta	Open-source, supports instrument and vocal generation

Leading GenAI models: Code generation

Model	Company	Use
GPT / GPT Turbo	OpenAI	Industry-leading coding assistant (used in Copilot and ChatGPT)
CodeWhisperer	Amazon	Developer-focused, AWS-integrated
Code LLaMA 70B	Meta	Open-source, good for research and experimentation

ChatGPT

ChatGPT is your AI-powered assistant developed by OpenAI, built to help you think, create, and solve problems in everyday life. It's thoughtful, useful, and conversational — whether you're exploring ideas, learning something new, or just getting things done.

Here are some of the key ways people use it:

- **Writing and communication:** ChatGPT can help draft emails, essays, stories, resumes, or social posts — and edit or refine your tone, grammar, and structure.
- **Learning and research help:** ChatGPT can explain difficult concepts, summarise long articles, or walk you through complex topics like math, history, or science.
- **Code assistance:** Whether you're learning to code or debugging complex programs, ChatGPT can help write, explain, or fix code across many languages.
- **Creative thinking:** Need ideas for a novel, brand name, birthday gift, or D&D campaign? ChatGPT can brainstorm with you.
- **Everyday tasks and productivity:** ChatGPT can help summarise documents, analyse spreadsheets, create schedules, or answer questions quickly.

Whether you need clarity, creativity, or just a bit of curiosity sparked, ChatGPT adapts to your style and goals.

Source: Copilot

Copilot

Copilot is your AI companion developed by Microsoft, designed to assist, inspire, and simplify tasks across your digital life. It’s helpful, creative, and collaborative — whether you're brainstorming, learning, writing, planning, or just exploring new ideas.

Here are some of the key ways people use it:

- **Writing and editing:** Copilot can help craft emails, essays, poems, or social media posts — and give helpful suggestions along the way.
- **Learning and research:** Ask Copilot anything from history to science, or help breaking down complex topics into digestible insights.
- **Coding support:** Copilot can assist with programming tasks in various languages, explain tricky bugs, or help generate scripts.
- **Idea generation:** From business names to weekend plans, Copilot is full of creative suggestions.
- **Everyday productivity:** Copilot can help plan your day, summarise long documents, or assist with spreadsheets and data analysis.

Copilot is built to adapt to your tone and needs — whether you want it serious, curious, or a little witty.

QA Source: Copilot

ChatGPT vs Copilot





ChatGPT:

- General purpose bot
- Conversational flair
- Great for creating writing
- Handles Q&A very well
- Supports multimodal input and output like image and audio

Copilot:

- Can be used on web for general tasks
- Built for productivity
- Embedded into Office apps, like Excel and Word
- Good for summarising documents, analysing spreadsheets, and writing documents
- Great for MS workflow

Prompt engineering

Prompt engineering is the art and science of crafting effective prompts or inputs to ask of GenAI tools (like ChatGPT or Copilot) so that it generates accurate, useful, and valuable outputs back to the user.

The important thing to remember is that it's not just asking questions, it's the ability to know how to ask the question in a way that gets you what you need.

The better you can explain what is required, the better the results.

Prompt engineering: Bad and good prompts

Bad prompts:

‘Fix my formula.’

‘Make me a chart from this data.’

Good prompts:

‘My formula is returning a #NAME error. What’s wrong and how to fix it?’

=IF(A1>100, A1*0.1, A1*0.05)’

‘Make me a chart from this data, showing total profit from column D by region from column.’

Prompt engineering: Application

Full sentences with ‘please’ and ‘thank you’ are not necessary.

It’s useful to use appropriate prompts to get better results:

- **Explain:** ‘Explain how to use XLOOKUP to match product name with ID…’
- **Generate:** ‘Generate an Excel formula to calculate profit and loss…’
- **Transform:** “Transform this messy data by removing duplicates and blanks…’
- **Troubleshoot:** ‘My Excel formula gives me a #VALUE error, troubleshoot this….’
- **Compare:** ‘Compare the use of Excel and Power Query for cleaning big datasets. Which is better for speed, efficiency, and ease of learning?’

Prompt engineering

Consider the scenario below:

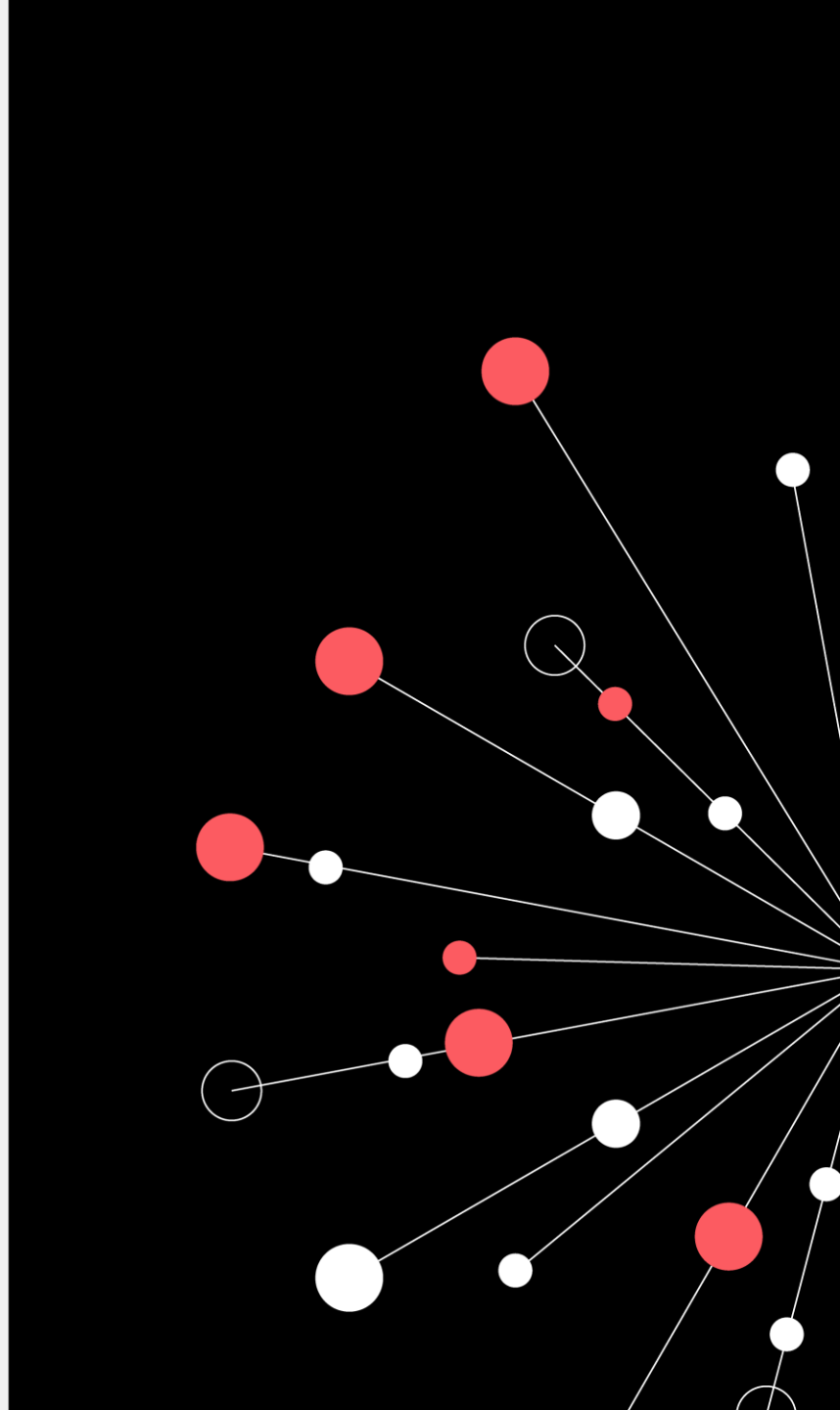
You are working on a large dataset, and you are unsure on how to clean it.

You want help from a GenAI tool.

You want to describe your requirements in plain English and get a response on how to clean the data in Excel steps, or Power Query Editor steps but not in VBA or Python code.

A prompt is the actual instruction you enter into the GenAI tool.

What would be an appropriate but concise prompt?



QA

Activity: AI prompts

Working in small groups, complete the task using one of the datasets (assigned by trainer).

Open files

"AI Prompts Task Sheet.docx"

Access to files (one per group)

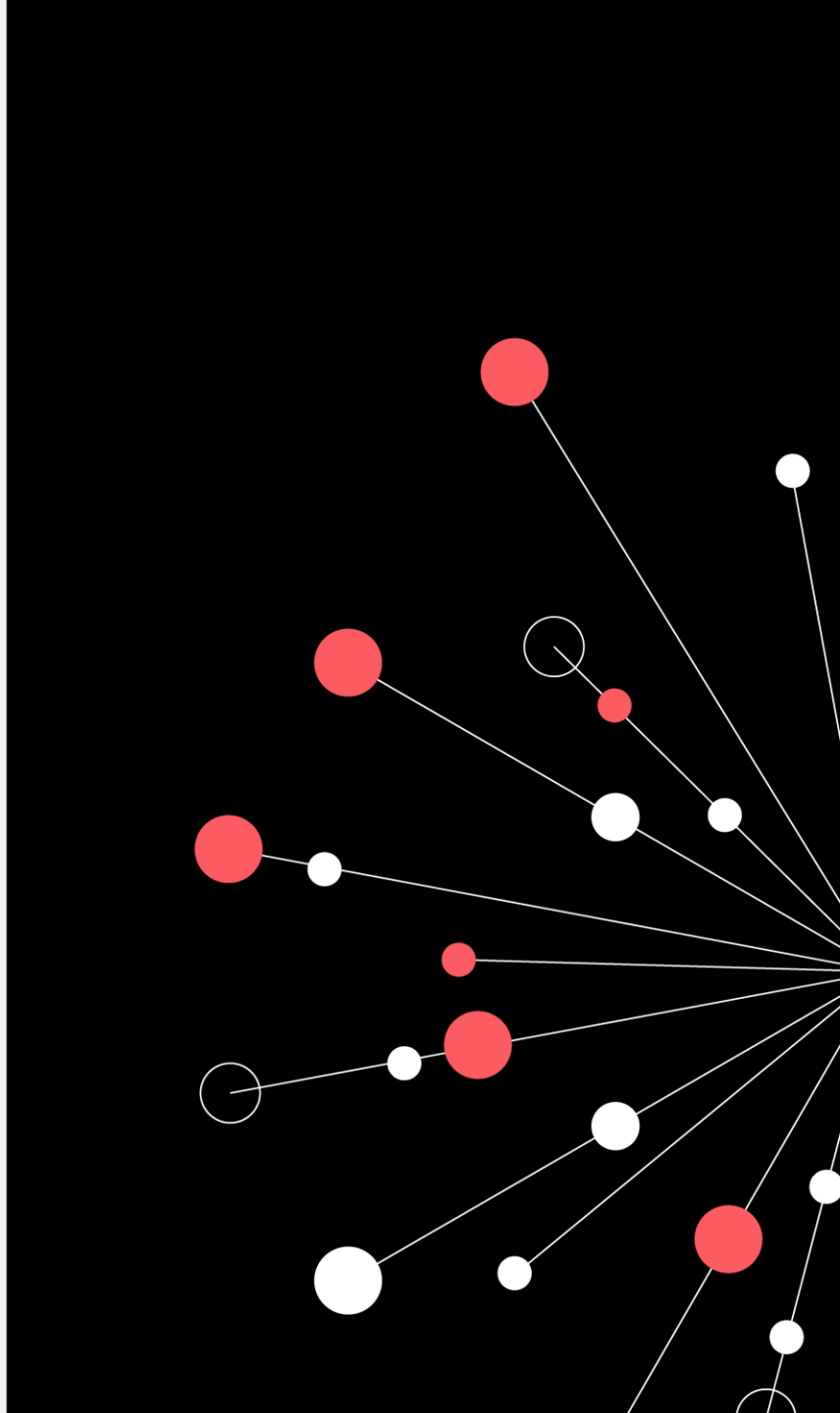
"Data Customer_Survey.xlsx"

"Data Employee_Performance.xlsx"

"Data Product_Inventory.xlsx"

"Data Sales_by_Region.xlsx"

"Data School_Attendance.xlsx"



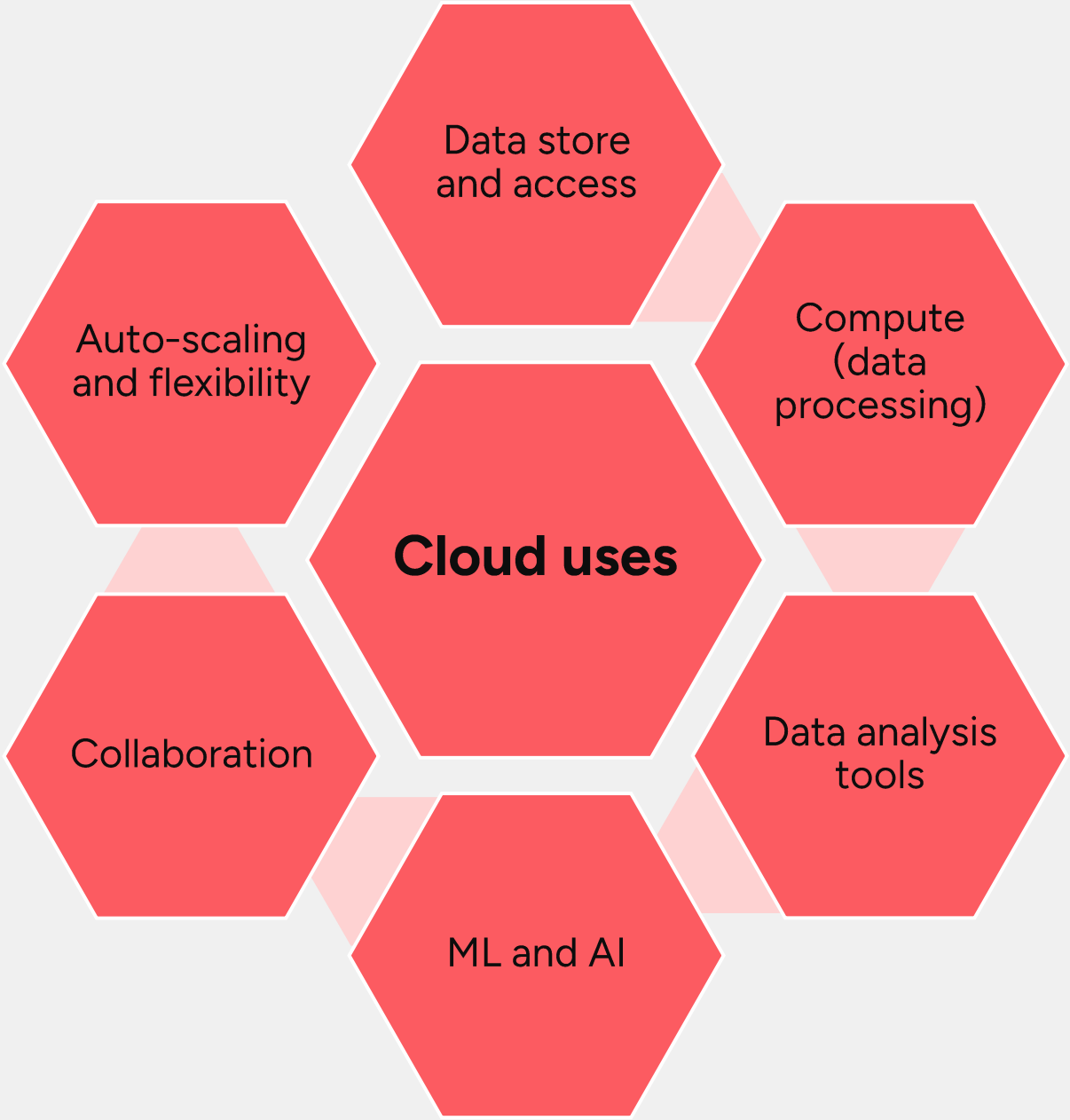
Cloud



Cloud

- Servers
- Accessed through the internet (through software and databases)
- Store, access, and process data
- Access software
- On-demand
- Broad device access
- Rapid elasticity
- Measured service
- Resource pooling
- Sustainability

Cloud: Key data uses



Cloud: Top providers

AWS		
Amazon platform. Offers storage, computing, databases, AI/ML, analytics.	**Tools for storage, virtual servers, ETL, data warehousing, ML.**	Best for: scalability, big data, enterprise-level analytics.
MS Azure		
Microsoft platform. Integrates well with Windows and Office tools.	**Tools for storage, cloud servers, data warehouse and analytics, Power BI.**	Best for: businesses using Microsoft products, enterprise IT, hybrid environments.
GCP		
Google platform. Known for strong data and AI tools, plus ease of use.	**Tools for fast data analysis on big data, notebooks, ML.**	Best for: data scientists, analytics-heavy work, cost-effective ML, Google companies

Cloud services: Organisational uses

In small groups, discuss and make notes on how your organisation can make use of cloud services:

- Which services would you recommend?
- How would that benefit your role / organisation's processes?

Feedback to the class.

